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Review

# Vehicle Brake Wear Particles: Formation Mechanisms, Behavior, and Health Impacts with an Emphasis on Ultrafine Particles

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**Abstract**

Brake wear particles (BWPs) represent a major source of non-exhaust particulate matter from road traffic, contributing substantially to human exposure, particularly in urban environments. While traditionally associated with coarse and fine fractions, mounting evidence shows that brake systems emit large quantities of ultrafine particles (UFPs; <100 nm), which dominate number concentrations despite contributing little to mass. This paper synthesizes current knowledge on BWP formation mechanisms, physicochemical characteristics, environmental behavior, and toxicological effects, with a specific emphasis on UFPs. Mechanical friction and high-temperature degradation of pad and disc materials generate nanoscale primary particles that rapidly agglomerate yet retain ultrafine structural features. Reported real-world and laboratory number concentrations commonly range from $10^3$ to over $10^6$ particles/cm$^3$, with diameters between 10 and 100 nm, rising sharply during intensive braking. Toxicological studies consistently demonstrate that UFP-rich and metal-laden BWPs, particularly those containing Fe, Cu, Mn, Cd, and Sb compounds, induce oxidative stress, inflammation, mitochondrial dysfunction, genotoxicity, and epithelial barrier disruption in human lung and immune cells. Ecotoxicological studies further reveal adverse impacts across aquatic organisms, plants, soil invertebrates, and mammals, with evidence of environmental persistence and food-chain transfer. Despite these findings, current regulatory frameworks address only the mass of particulate matter from brakes and omit UFP number-based limits, leaving a major gap in emission control.

**Keywords:** brake wear particles; braking systems; ecotoxicological impact; exposure; health impact; toxicity; ultrafine particles; vehicle emissions

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## 1. Introduction

Road transport is one of the main sources of air pollution in urban areas [1,2]. The growing number of vehicles adversely affects air quality, particularly through the emission of particulate matter (PM), carbon monoxide (CO), nitrogen dioxide ($NO_2$), and ozone ($O_3$) [2]. Particulate matter (PM) is a pollutant of concern not only because of its harmful effects on human health but also due to its capacity to reduce visibility and contaminate buildings and materials [3]. In general, PM is classified according to particle diameter (dp) into coarse ($PM_{10}$, 2.5 μm < dp < 10 μm), fine ($PM_{2.5}$, dp < 2.5 μm), and ultrafine ($PM_{0.1}$, dp < 0.1 μm) fractions [4,5]. Traffic-related emissions are dominated by ultrafine particles

(UFPs), and globally, road transport represents the primary source of UFP concentrations in outdoor urban air [6].

Traffic emissions are generally classified into two categories: exhaust and non-exhaust emissions. Non-exhaust sources primarily include particles generated from brake wear, tire wear, and road surface abrasion [7].

Particles generated from brake wear cover a broad size range—from coarse (2.5–10 μm) and fine (0.1–2.5 μm) to ultrafine (<0.1 μm), extending down to a few nanometers [8]. Within brake wear emissions, both discrete and agglomerated UFPs dominate [9]. Agglomerated particles are composed of clusters of primary (discrete, individual) particles [10]. According to Gonet et al. [9], the larger fractions of airborne brake wear emissions are mainly composed of ultrafine agglomerates. Typical urban PM size distributions exhibit multiple peaks or modes. For UFPs, two peaks are commonly observed in the number distribution: one corresponding to the nucleation mode (particles < 0.01 μm) and another to the Aitken mode (particles between 0.01 μm and 0.1 μm) [11].

In the literature, the terms "brake wear debris", "brake wear dust", "brake wear emissions" and "brake wear particles" are used interchangeably to describe particles generated from the wear of vehicle braking systems. These particles differ in morphology, chemical composition, and most importantly, size. Although braking processes are predominantly mechanical and thus expected to generate particles mainly within the coarse fraction, several studies have reported substantial concentrations of brake-derived particles in both the fine and ultrafine size ranges [12–14]. For clarity and consistency, the term "brake wear particles" and its abbreviation "BWPs" are used throughout the following text.

BWPs are considered a significant [11,12,15] or the most important [16] source of non-exhaust PM emissions from motor vehicles. In European Union (EU) countries, BWPs account for approx. from 8 to 27% of total traffic-related emissions [17]. Most airborne BWPs (>99% of particle concentration) are smaller than 200 nm. Transmission electron microscopy (TEM) analysis have confirmed that even larger fractions of airborne BWPs predominantly consist of UFP agglomerates [9].

Exhaust emissions are strictly regulated; vehicle manufacturers must comply with stringent emission standards, and most diesel vehicles on the road today are equipped with particulate filters [18]. In contrast, non-exhaust emissions remain unregulated and, in addition to BWPs, typically include particles generated by tire abrasion and resuspended road dust [7]. By 2050, non-exhaust emissions are expected to constitute up to 90% of all PM emitted by road traffic due to the reduction in exhaust emissions from vehicle electrification. Therefore, effective monitoring and regulation of non-exhaust emissions are crucial [19].

In terms of composition, UFPs comprise both heterogeneous and homogeneous core particles, solid carbon aggregates, and trace metal compounds. These structures may contain from a single particle to several hundred individual particles, along with adsorbed and absorbed semi-volatile and volatile compounds [20]. UFPs containing iron (Fe) can be toxic by themselves and are often associated with other transition and heavy metals such as chromium (Cr), copper (Cu), manganese (Mn), nickel (Ni), lead (Pb), and zinc (Zn). BWPs represent a major source of iron-containing UFPs in roadside air [9]. A significant fraction of solid UFPs emitted from brake systems can be toxic to the environment [21]. These UFPs are of particular environmental and health concern, as they can easily penetrate the human body through inhalation and contribute to the development of various severe diseases [22].

Previous studies have demonstrated both the potential and confirmed adverse effects of UFPs on the environment and human health. However, epidemiological evidence regarding UFP-related health impacts remains limited. Hypothesized health effects include increased cardiovascular and respiratory morbidity and mortality, induction of local pulmonary and systemic inflammation, oxidative stress, and adverse impacts on brain

function, the nervous system, and metabolism [23]. UFPs are also believed to pose risks to neuronal and cardiovascular health upon inhalation [9]. Their uptake into plant tissues may facilitate entry into food chains [15]. According to Richtmann et al. [15], metallic UFPs from brake wear play a significant role in eliciting physiological responses. Furthermore, key groups of volatile organic compounds, including polycyclic aromatic hydrocarbons (PAHs) and benzene, toluene, ethylbenzene and xylenes (BTEX) compounds, have been confirmed to form during braking [24], and BWPs have shown genotoxic properties [25]. These findings underscore the importance of developing a comprehensive understanding of UFP toxicity and their effects on human health.

The European Commission has revised the Air Quality Directive and recommended regular measurement of UFP concentrations at selected sites within EU member states [26]. The highest concentrations are typically observed near roads and transport hubs, where they can exceed background levels by up to an order of magnitude. Although particle numbers decrease through coalescence and diffusion processes, UFPs can remain suspended in the atmosphere for extended periods and be transported over considerable distances [27].

The generation and chemical composition of UFPs from brake wear depend on several factors, including braking style [28] and brake temperature [29,30]. Extremely high UFP concentrations, approaching 1,000,000 particles per cubic centimeter (hereafter expressed as $\#/cm^3$), often result from new particle formation in urban environments; however, the fundamental mechanisms governing UFP nucleation and growth remain poorly understood. The EURO 7 emission standards introduce limits for solid particle emissions from brakes at 7 mg/km for internal combustion vehicles and 3 mg/km for electric vehicles, along with a unified testing methodology under UN Global Technical Regulation No. 24. This framework enables manufacturers to better compare and optimize strategies to reduce brake-related particulate matter emissions. Nonetheless, EURO 7 does not establish limits for particles smaller than $PM_{10}$, particularly UFPs. This regulatory gap represents a critical deficiency in current traffic emission policies, which future research based on this paper could address through the proposal of suitable emission limits, mitigation measures, or brake system modifications [19,31,32].

Previous studies have investigated the mechanisms of particle emissions from brake systems, including UFPs, under various operating conditions. Research has examined emission characteristics as a function of temperature [22], brake disc material [21,32], brake pad composition [33], disc surface treatment [34], contact pressure [35,36], sliding velocity [35–37], and braking cycle parameters [38]. Furthermore, studies have explored the influence of different reinforcing fibers in brake pads on BWPs during braking [39], as well as the effects of braking conditions, specifically speed and deceleration on UFP generation [40].

Overall, there is currently a lack of a comprehensive overview regarding the mechanisms, behavior, and environmental impact of UFPs generated from brake systems. Therefore, the aim of this paper is to provide an integrated overview of BWPs, with particular emphasis on the UFPs. Current knowledge on the mechanisms governing UFP formation during braking, their physicochemical characteristics, environmental behavior, and documented ecotoxicological effects across biological systems is summarized in this study. Furthermore, this study evaluates methodological approaches used to measure BWPs, including UFPs, and highlights key regulatory gaps, particularly the absence of particle number–based emission limits for brake systems. By synthesizing these aspects, this study aims to support the development of improved monitoring strategies and more effective regulatory frameworks for non-exhaust traffic emissions.

## 2. Ultrafine Particles Characteristics and Relevance to Brake Wear

UFPs are sometimes referred to in the literature as nanoparticles due to their comparable size. However, the term "nanoparticles" is typically reserved within the scientific community for particles ≤ 100 nm that are intentionally produced through controlled engineering processes [28]. Dimensionally, both UFPs and nanoparticles fall within the same size range, with particle diameters < 0.1 µm. The key distinction lies in their origin. UFPs form spontaneously in the environment, often as by-products of fossil fuel combustion, condensation of semi-volatile compounds, or various industrial emissions, whereas nanoparticles are deliberately synthesized through precisely controlled technological methods. Compared to the extensive knowledge available on the health effects of particles with aerodynamic diameters below 10 µm (coarse PM) or 2.5 µm (fine PM), there remains a substantial gap in understanding the impacts of particles smaller than 100 nm. Growing evidence from air pollution research indicates that submicron particles exhibit physicochemical properties markedly different from those of larger PM fractions and may therefore exert distinct adverse health effects. These effects are particularly associated with respiratory diseases, asthma development, and allergic sensitization to common allergens through multiple mechanisms of action. Table 1 compares selected principal characteristics of $PM_{10}$, $PM_{2.5}$ and UFPs [41,42].

**Table 1.** Comparison of $PM_{10}$, $PM_{2.5}$, and UFPs.

| Parameter | $PM_{10}$ | $PM_{2.5}$ | UFPs |
|---|---|---|---|
| Particle diameter (µm) | 2.5–10 | 0.1–2.5 | <0.1 |
| Deposition in the alveolar space | no | no | yes |
| Ratio of surface area to mass | + | ++ | +++ |
| Organic carbon content | + | ++ | +++ |
| Elemental carbon content | +++ | ++ | + |
| Metal content | +++ | ++ | + |
| Measurement metric | mass concentration | mass concentration | particle number or surface area |

Note: The symbols +, ++, and +++ indicate the relative level of the given property, where + represents a low, ++ a medium, and +++ a high value compared to other particle categories.

Additional properties of UFPs and nanoparticles include manifestations of quantum phenomena, such as changes in optical, electrical, mechanical, and magnetic properties [41].

UFPs in the environment originate from both natural and anthropogenic sources. When formed during combustion processes, they typically have a core composed of elemental or organic carbon containing trace amounts of metals, sulfates, or volatile and semi-volatile components. Other components of combustion-derived UFPs depend on the type of fuel, combustion conditions, and atmospheric parameters. Given the nearly ubiquitous nature of UFP sources, their presence in outdoor and indoor air is neither novel nor unusual. Typical UFP concentrations in rural ambient air are 2610 particles #/cm$^3$, while concentrations near roadways can reach up to 48,180 #/cm$^3$. The global average concentration is 10,760 #/cm$^3$. The number of UFPs decreases rapidly due to coagulation into larger particles and atmospheric dispersion, leading to locally elevated concentrations near roads or industrial sites. Peak traffic-related emission concentrations occur near road edges and are often more than ten times higher than background levels. Concentrations generally return to background levels at approx. 500 m from the source. Although UFP numbers decrease quickly due to coagulation, these particles remain airborne for extended periods and are subject to transport processes [27].

High UFP concentrations in the air are associated with various conditions, including high humidity, low air flow diffusion, increased numbers of diesel vehicles, seasonal variations, frequent vehicle acceleration after stops, braking, and tire wear on road surfaces. Improvements in fuel and engine technologies and the use of catalytic converters have reduced the amount of PM and carbon monoxide (CO) from vehicle exhaust, but have, conversely, increased UFP number and toxicity. It is also important to note the increasing number of electric vehicles which, although they produce no exhaust emissions, contribute significantly more to non-exhaust PM emissions than conventional internal combustion vehicles due to their higher weight—by up to 24% more [43].

Epidemiological and clinical studies indicate that UFPs may be substantially more hazardous than $PM_{10}$ and $PM_{2.5}$ due to their chemical composition, small size, high surface area-to-mass ratio, ability to generate reactive oxygen species (ROS), high retention rates, and capacity to penetrate deeply into the respiratory system. Clinical findings have shown that UFP exposure is associated with reduced peak expiratory flow and increased respiratory symptoms in individuals with asthma. Emerging evidence also points to harmful cardiovascular effects, including alterations in heart rate, impaired microvascular function, and systemic inflammation. A positive association has been reported between UFP exposure and ischemic heart disease mortality in a cohort of 100,000 subjects. Inhalation exposure has been shown to cause acute adverse cardiopulmonary effects, such as reduced arterial oxygen saturation in both healthy and asthmatic individuals. Moreover, a single two-hour UFP inhalation was found to disrupt heart rate and heart rate variability in diabetic subjects [44–52].

Several key facts highlight the critical importance of addressing the adverse health effects of UFPs [41]:

- Although $PM_{10}$ and $PM_{2.5}$ can be readily removed by phagocytosis, the extremely small size of UFPs allows them to evade such host defenses and deposit in the lungs with a high retention rate. For the same volume of inhaled air, the actual dose and local effects of UFPs in the lungs may therefore be substantially greater than for $PM_{2.5}$. Furthermore, UFP size facilitates translocation to other organs via systemic circulation, leading to toxicological mechanisms that are markedly different from those of $PM_{2.5}$;
- The large surface area of UFPs enables them to carry significant amounts of adsorbed hazardous substances, including organic chemicals and metals, which can generate ROS and oxidative stress. Oxidative damage plays a critical role in the adverse health effects induced by UFPs, particularly in asthma, chronic obstructive pulmonary disease (COPD), and atherosclerosis;
- Unlike coarse and fine particulate matter, UFPs are not evenly distributed in the atmosphere but are instead concentrated near emission sources, such as heavily trafficked roads. This localized distribution has resulted in limited monitoring infrastructure and a scarcity of large-scale epidemiological studies—conditions unlikely to improve until regulatory authorities begin measuring UFPs as criterion pollutants;
- The composition of semi-volatile organic compounds on the surface of UFPs can dynamically change depending on the source and molecular size, complicating efforts to draw simple conclusions regarding their health impacts;
- The effects of $PM_{10}$ and $PM_{2\cdot5}$ are typically evaluated based on their mass; however, UFPs contribute negligibly to total mass while dominating the atmospheric particle number. This necessitates the use of alternative exposure metrics, such as particle number concentration. Epidemiological studies employing these metrics remain relatively scarce.

According Liati et al. [8], UFPs are a critical fraction of BWPs because they can dominate particle number concentrations even when contributing relatively little to total mass,

and their small size increases the probability of deep lung deposition and interaction with biological tissue. Brake wear generates particles spanning micro- to ultrafine-scales, and electron microscopy-based studies have confirmed the significant presence of UFPs in BWPs with complex morphologies and mixed inorganic composition, supporting the view that brake systems are an important source of traffic-related UFPs. From a toxicological perspective, UFPs are of particular concern because BWPs are often enriched with reactive components that can drive redox activity and oxidative injury.

## 3. Methodology

### *3.1. Selection and Distribution of Published Studies on Brake Wear and Ultrafine Particles*

This paper was conducted using a structured and comprehensive literature search strategy to capture current knowledge on the generation, physicochemical characteristics, environmental behavior, and toxicological impacts of BWPs, with a particular emphasis on UFPs. PRISMA 2020 guidelines, which provide a standardized approach for identifying, selecting, and critically appraising relevant studies, as well as collecting and analysing data were used [53].

Scientific publications were identified in the PubMed, Google Scholar, Scopus and Web of Science databases (accessed on 1 May 2025) using combinations of relevant keywords, including "brake wear particles"; "brake wear debris"; "brake wear dust"; "ultrafine particles"; "non-exhaust emissions"; "vehicle emissions"; "solid particles"; "toxicity"; "ecotoxicity"; "inhalation;" "exposure" and "environmental health". Studies were included based on the following predefined eligibility criteria:

- Study Design: Original research articles, including experimental studies and reviews.
- Language: Articles published in English.
- Publication Date: Articles published up to 2025, with particular emphasis on the period 2015–2025.
- Country of origin: Irrelevant.
- Outcomes: Each reference was classified into one of the following three categories:
    - Articles primarily focused on BWPs and their environmental and/or health impacts.
    - Articles primarily focused on UFPs, particle number concentrations, their physicochemical properties, health effects, or exposure assessment, including articles without an explicit focus on braking systems.
    - Overlapping articles addressing PM, BWPs, and UFPs in a general context, including regulatory, toxicological, epidemiological, or methodological aspects, were identified but not included in either of the two target categories. If an articles addressed BWPs with a specific emphasis on UFPs, it was counted in both categories, which is methodologically justified as it directly addresses both research themes.
- Exclusion Criteria: Editorials, conference abstracts, and studies not reporting any of the above outcomes were excluded.

Both original research articles and reviews were screened, but only primary research articles reporting experimental or observational data were used as core sources for data synthesis. Articles collected by initial search were filtered based on title, abstract, and methodological relevance, with particular attention given to studies reporting particle size distributions, mass or number concentrations, emission mechanisms, toxicological and ecotoxicological endpoints. Relevant review articles were additionally examined to identify further sources not captured in the initial search. The final dataset included studies

spanning in vitro, in vivo, ecotoxicological, and exposure-assessment methodologies, as well as tribological and physicochemical analyses of BWPs and UFPs, respectively.

Particular emphasis was placed on extracting data relevant to UFP generation, size of the particles, number-based emission metrics, and metal-associated toxicity, reflecting their central role in health and environmental impacts. This structured approach enabled a coherent synthesis of multidisciplinary findings and provided a comprehensive evidence base for identifying mechanistic pathways, exposure risks, and critical gaps in existing regulatory frameworks. The methodology thus supports the overarching aim of the paper: to offer an integrated assessment of BWPs, with a focus on UFPs and their implications for environmental and human health.

Figure 1 presents the PRISMA 2020 flow diagram showing the number of articles analysed in this study addressing BWPs and UFPs.

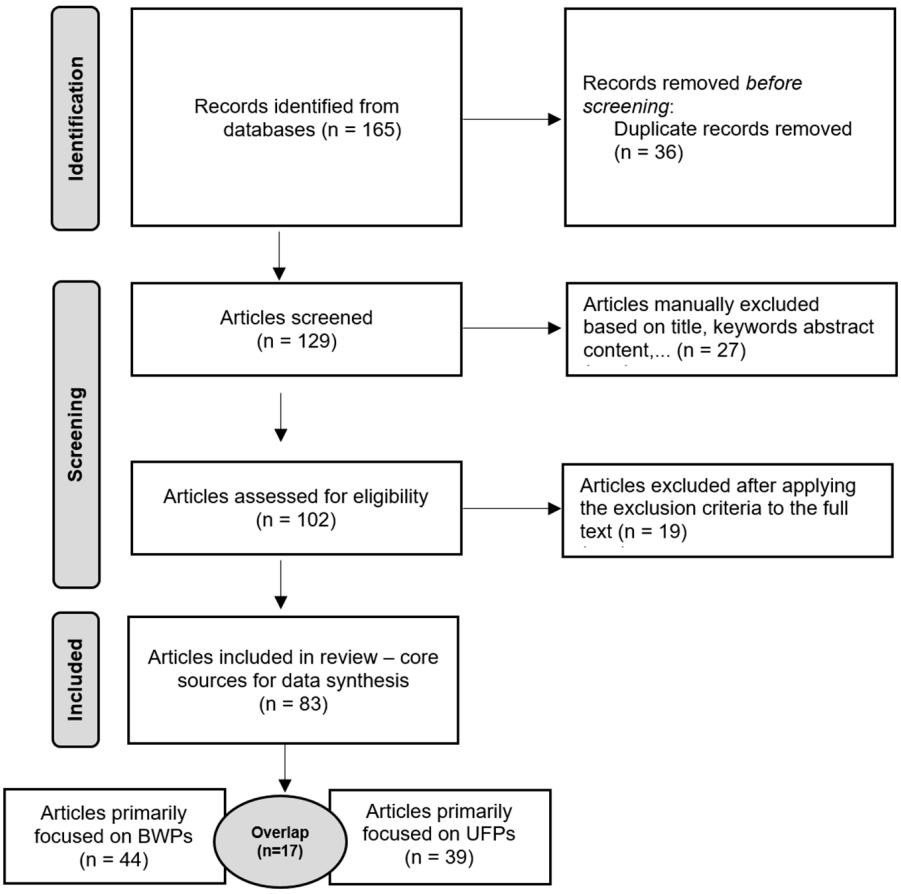


**Figure 1.** PRISMA 2020 flow diagram illustrating identification, screening, eligibility assessment, and thematic inclusion of studies addressing brake wear and ultrafine particles.

The regional distribution of these publications is dominated by Europe (52%), followed by East Asia (26%) and North America (14%). Europe primarily leads research on BWPs and regulatory-oriented studies, whereas East Asia shows particular strength in experimental brake tribology and emission characterization. North America predominantly contributes to health and toxicological studies focusing on UFPs. From a temporal perspective, a strong increase in the number of publications has been observed since 2015, with a marked acceleration from 2019 onwards. This trend reflects increasing regulatory pressure and growing concern regarding non-exhaust emissions and UFPs. Overall, 76% of the articles analysed in this study were published after 2015, and 49% were published from 2020 onwards.

### 3.2. Use of Artificial Intelligence Tools in Data Visualization

Figures 2 and 3 were generated with the assistance of generative artificial intelligence (GenAI) tool, ChatGPT (model 5.2). GenAI was used exclusively for the creation and visual rendering of graphical elements based on author-defined inputs and specifications. GenAI tools were not used for data generation, data analysis, or interpretation of the study results.

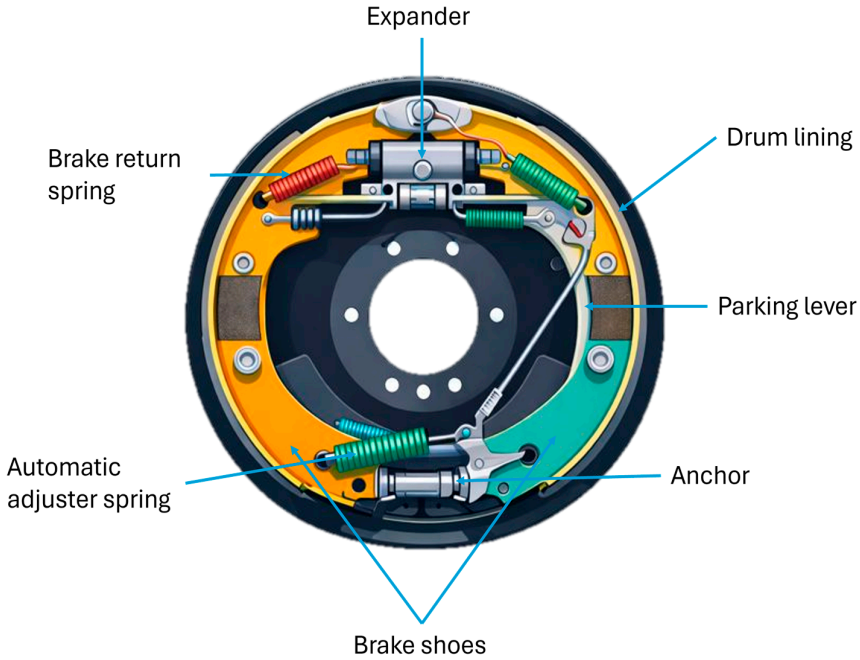


**Figure 2.** Illustration and description of a drum brake.

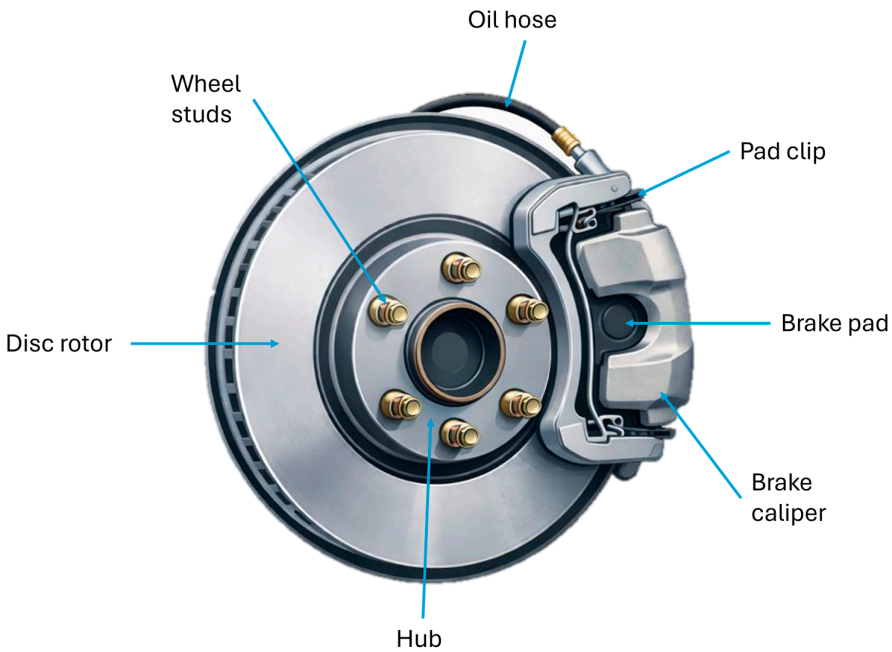


**Figure 3.** Illustration and description of a disk brake.

## 4. Brake System Design and Mechanisms of Brake Wear and Ultrafine Particles Generation

The evolution of vehicle brake systems is closely associated with increasing traffic speed and density, both of which demand the ability to stop vehicles within short distances. The primary function of a vehicle's braking system is therefore to decelerate or bring

the vehicle to a complete stop. In the earliest automobiles, braking did not pose a major challenge. These vehicles operated at relatively low speeds and used brakes acting on the external surface of the wheel, employing movable shoes or friction linings similar to those found on horse-drawn carriages and railway cars. However, with the introduction of air-filled rubber tires, this simple design became inadequate due to excessive wear [54].

Currently, two main types of brake systems are used in road vehicles: drum brakes and disc brakes. The purpose of this study is not to provide a detailed technical description of these systems but to outline their fundamental characteristics relevant to the context of the present research.

Drum brakes (see Figure 2) were the first type of braking system to be developed and were initially applied only to the rear wheels. Although more complex to manufacture, they provided effective heat dissipation during prolonged braking. Band and shoe brakes were not entirely replaced following the introduction of drum brakes but continued to serve primarily as parking (hand) brakes in automobiles. The principal component of the drum brake is the brake drum, which is rigidly attached to the wheel and rotates with it. Within the drum, brake shoes fitted with friction linings are positioned. When the driver depresses the brake pedal, a hydraulic, mechanical, or electronic actuator forces the shoes outward, pressing them against the inner surface of the drum. The resulting friction reduces or halts the drum's rotation, thereby decelerating or stopping the wheel [55].

Although the British automobile manufacturer Lancaster patented the disc brake as early as 1902, its practical implementation was significantly delayed. Disc brakes (see Figure 3) were first introduced into mass production by Jaguar in 1953 [54] and later by Citroën in 1955 with the iconic DS-19 model [56]. The fundamental components of a disc brake system include the brake disc, brake pads, and brake caliper.

The brake disc is a metal plate that rotates together with the wheel. During braking, the brake pads are pressed against the disc surface, generating friction that slows the vehicle. The pads are housed within the brake caliper, which contains pistons actuated by hydraulic pressure. This pressure is produced when the brake pedal is depressed and transmitted through the brake fluid. Brake pads are composed of friction materials capable of withstanding high temperatures and pressures, and they may be organic, semi-metallic, ceramic, or metallic in composition. The appropriate selection of brake pad materials influences not only braking efficiency but also noise generation and component wear. Compared with drum brakes, disc brakes provide superior braking performance, greater thermal stability, and faster response times [57].

Brake pads typically comprise five principal components: binders, fibers, fillers, additives or lubricants, and abrasives. The binder serves to hold the composite constituents together and to maintain structural integrity under mechanical and thermal stress. It generally constitutes 20–40 wt.% of the pad material and is most commonly composed of modified phenol–formaldehyde resins. Reinforcing fibers provide mechanical strength and structural stability, typically representing 6–35 wt.% of the pad composition. These fibers are commonly made from copper, steel, brass, potassium titanate, glass, organic materials, or aramid fibers such as Kevlar [57].

Fillers are incorporated to improve the thermal stability and noise-dampening properties of the brake pad while also reducing production costs. They typically consist of inorganic compounds such as barium (Ba) and antimony trisulfide ($Sb_2S_3$), magnesium (Mg) and chromium oxides, silicates, ground slag, stone, and metal powders, accounting for approx. from 15 to 70 wt.% of the pad material. Lubricants influence the wear behavior of the pad and may be inorganic, metallic, or organic in nature. Graphite is most commonly used, although other materials such as ground rubber, metal particles, soot, and antimony trisulfide are also employed. Lubricants typically constitute 5–29 wt.% of the brake pad.

Abrasives are added to enhance friction, maintain cleanliness between contact surfaces, and prevent excessive buildup of transfer films. They usually represent about 10 wt.% of the pad composition and commonly include aluminum oxide, iron oxides, quartz, and zircon. The proportions of these components vary depending on the pad type, manufacturer, and vehicle application [56,57].

In modern passenger vehicles, three primary types of brake pads are commonly used: non-asbestos organic (NAO), semi-metallic, and low-metallic pads [57]. Both drum and disc brake systems require suitable friction materials, and asbestos was historically considered particularly effective for this purpose. However, after asbestos fibers were identified as hazardous to human health, they were replaced by alternative materials [56]. Non-asbestos organic pads are relatively soft and produce low braking noise compared with other pad types but exhibit reduced braking efficiency at high temperatures and generate higher levels of dust emissions. Low-metallic pads consist of organic compounds blended with a small proportion of metal (10–30 wt.%), providing high friction and reliable performance under elevated temperatures. Semi-metallic pads contain a higher metal content (up to 65 wt.%), which improves durability and thermal conductivity; however, they are prone to faster wear and increased noise generation during braking [57].

Braking is a process in which a vehicle's kinetic energy is transformed into thermal energy through friction between the brake pads and discs, or between the brake shoes and drums, depending on the braking system design. This frictional contact results in mechanical wear of the friction surfaces and the subsequent release of particles of varying sizes and compositions, predominantly containing elements such as Fe, Cu, Pb, Zn, calcium (Ca), antimony (Sb), tin (Sn), barium (Ba), and silicon (Si), respectively. The material composition of the brake friction surfaces significantly influences both the particle size distribution and the concentration of emissions generated during braking [58].

UFPs originating from braking processes are generated primarily through two mechanisms: mechanical friction and thermal degradation of materials. Mechanical friction in the braking system arises from the physical contact and intense pressure between two surfaces of differing hardness and composition. This interaction induces microstructural damage of the friction surfaces, including [58]:

- Formation of microcracks and surface deformations on the friction material;
- Progressive detachment and separation of microparticles from the material surface, followed by their mechanical fragmentation into smaller particles;
- Further disintegration of these fragments under mechanical energy, producing particles that are released into the air as primary UFPs.

According to Straffelini and Gialanella [11], coarse and fine particles are largely produced by abrasion and detachment of primary wear debris, the UFPs are frequently associated with thermal events in the contact (flash temperatures, hot spots) that promote material transformation, oxidation, and release of semi-volatile/volatile species that subsequently condense.

During braking, friction between the brake pad and disc generates substantial heat, with temperatures reaching from 500 to 700 °C and, under extreme conditions, even higher. At such elevated temperatures, the thermal decomposition (pyrolysis) of organic components within the brake materials occurs. In addition to metallic and ceramic constituents, brake pads contain organic binders, resins, and polymers that contribute to structural strength and durability. When exposed to high temperatures, these organic compounds decompose into smaller molecules, forming volatile organic compounds such as aldehydes, ketones, aromatic hydrocarbons, and other complex organics. Upon rapid cooling near the braking surfaces, these gases and vapors condense onto existing particles, leading to the formation of secondary UFPs. The quantity and composition of the resulting parti-

cles depend on multiple factors, including the materials used in the brake components, braking intensity, vehicle speed, ambient conditions, and the wear state of the braking system [59–61]. In addition Woo et al. [62] showed, that when pad temperature exceeds a critical threshold during repeated harsh braking, high number concentrations of volatile UFPs in the range from 10 to 50 nm range are attributed to volatilization and subsequent nucleation, with particle growth occurring by condensation and coagulation in the cooling airflow downstream of the contact. Niemann et al. [59] concluded, that UFPs can increase markedly as the friction interface approaches elevated temperature regimes, consistent with a transition from predominantly mechanical wear to thermally driven particle formation.

## 5. Methodologies for the Measurement of Brake Wear and Ultrafine Particles

### *5.1. Sampling Techniques Overview*

BWPs are generated through mechanical abrasion and thermal processes occurring at the pad–disc or pad–drum interface during braking. Due to their heterogeneous composition, broad size distribution, and rapid dispersion, their accurate measurement requires carefully designed sampling strategies [63].

Sampling techniques are commonly divided into laboratory-based and field-based approaches. Laboratory-based measurements dominate the literature and typically employ brake dynamometers, chassis dynamometers, inertial dynamometers, or tribometers. These setups enable precise control of braking parameters such as speed, braking force, temperature, and braking cycle, ensuring high reproducibility and allowing BWPs to be isolated from other non-exhaust sources. Chassis dynamometers additionally permit testing of complete vehicles under standardized driving cycles, bridging the gap between component-level testing and real-world operation. Field-based techniques include on-vehicle sampling systems, roadside measurements, and tunnel studies, which capture brake wear aerosols under real traffic conditions. While these approaches provide valuable information on actual exposure levels, they are subject to higher uncertainty due to meteorological influences, traffic variability, and challenges in source apportionment [63].

Brake wear aerosols are collected using several complementary mechanisms. Gravimetric filter sampling remains a foundational method for determining particulate mass and for subsequent chemical and morphological analyses. However, its limited temporal resolution has led to widespread use of real-time aerosol instruments, including optical particle counters (OPCs), condensation particle counters (CPCs), aerodynamic particle sizers (APS), and electrical mobility-based instruments such as scanning mobility particle sizers (SMPS), fast mobility particle sizers (FMPS), and engine exhaust particle sizers (EEPS) [63].

Size-segregated sampling is commonly achieved using cascade impactors, nozzle impactors, or electrical low-pressure impactors (ELPI), which allow particles to be classified according to aerodynamic diameter. These techniques are particularly important given the strong influence of particle size on atmospheric transport and respiratory deposition. Resonance-based instruments, such as tapered element oscillating microbalances (TEOM) and quartz crystal microbalances (QCM), are used for real-time mass measurements but are typically combined with other instruments due to their limited sensitivity to UFPs [63].

Also. sampling strategies differ in temporal resolution, ranging from off-line methods, such as SEM-EDX, ICP-MS, and ICP-AES, to real-time monitoring systems that provide continuous particle number and size data. Most studies combine real-time measurements with off-line chemical analyses to identify brake-specific elemental markers, including Cu, Fe, Sb, and Ba. This multimodal approach is increasingly recognized as necessary, as no single technique can fully characterize the physical and chemical properties of BWPs [63].

For the measurement of UFPs, recommended techniques are electrical mobility–based instruments (SMPS, FMPS, EEPS) and CPCs, as they provide the required sensitivity for the UFP size range and, in the case of mobility analyzers, size-resolved number distributions suitable for transient braking conditions. ELPI can partially capture UFPs while enabling concurrent filter collection for chemical analysis, but its lower-nanometre performance is more sensitive to operational settings. In contrast, not recommended as standalone methods for UFP assessment are optical instruments (OPC, photometer), APS, and mass-based techniques (filter gravimetry, TEOM, QCM), which either lack sufficient sensitivity below approx. 200 nm or inherently underrepresent UFPs due to their negligible mass contribution. Accurate UFP characterization therefore requires mobility-based sizing and/or CPCs, preferably within integrated, multimodal sampling setups [63].

### *5.2. Methodological Approaches for Sampling and Exposure Assessment of Brake Wear and Ultrafine Particles in Published Studies*

When evaluating the BWPs, attention must be paid to the methodology employed in the studies. The main methods used for sampling BWPs encompass a wide range of laboratory-based and field-representative techniques, each with specific advantages and limitations. Among these, the brake dynamometer is the most commonly applied tool for generating BWPs under controlled and repeatable braking conditions [15,25,64–73]. Dynamometers enable simulation of diverse braking scenarios, from mild deceleration to extreme full-stop events, and produce sufficient quantities of particles for physicochemical characterization and toxicological testing. However, depending on the selected braking protocol, dynamometer-generated samples may not always accurately reflect real-world emissions, particularly when elevated temperatures or excessive braking forces alter particle morphology or chemical composition.

Another frequently used approach involves filters installed in the ventilation duct of brake testing equipment. These filters capture airborne particles emitted during braking over extended periods, providing representative mixtures of particles accumulated under routine operational conditions [74]. Meanwhile, collection of particles directly from brake drums allows sampling of wear debris that accumulates within enclosed braking systems, although such samples are often coarser and may not fully represent respirable fractions relevant for inhalation exposure [75,76].

Similarly, particles collected from the friction material of brake pads provide insights into the source composition of brake-derived particles. These samples are particularly useful for elemental and mineralogical analysis, especially when investigating the contribution of pad formulations to toxicity endpoints such as metal-induced oxidative stress [77–79].

High-volume cascade impactors and other impactor systems permit size-segregated sampling of airborne BWPs across defined aerodynamic diameter ranges, enabling targeted toxicological studies on fractions such as $PM_{2.5}$ or UFPs [80,81].

Exposure chambers are used when direct exposure of living cells or organisms to freshly generated BWPs is required, providing controlled environments with regulated airflow, temperature, and humidity [82].

Time-integrated sampling devices, such as time-integrated BWPs collection systems, accumulate particles over extended periods and allow quantification of long-term exposure patterns. In contrast, field resuspension chambers simulate particle mobilization from road surfaces under natural conditions, thereby enabling the study of resuspended BWPs that contribute to ambient PM levels in urban environments [80,83].

Finally, more unconventional methods such as grinding of brake shoes are employed to obtain large quantities of particulate material for testing. While efficient for generating bulk samples, this method may produce particles with physicochemical properties that

differ from those emitted during actual braking events, thus requiring careful interpretation of results [84].

In some ecotoxicological studies, instead of sample collection, BWPs exposure was measured directly or estimated through data modeling. Based on our analysis, the following approaches were identified:

- Randomized two-factor crossover exposure analysis [85], randomized crossover design [44], and double-blind randomized crossover study [50,51];
- Rotational factor analysis with maximization of variance [86];
- Analysis of total particle number concentration using a UFP detector with mobile scanning instruments and particle counter [45,48,87];
- Analysis of concentration data obtained at the institutional level [88] and from monitoring stations [46,89,90];
- Exposure estimation using a chemical-transport models [49];
- Direct exposure of organisms, such as forager bees [91], pregnant female mice [92], or male rats [73,84].

From our perspective, in ecotoxicological studies it is essential to distinguish whether BWP samples were first collected and subsequently subjected to toxicity testing, or whether organisms were directly exposed to BWPs, with exposure levels potentially estimated through modeling.

In the first approach, a brake dynamometer was again the most commonly used tool to simulate real braking cycles during BWP collection. Puisney et al. [64] noted, that extreme conditions are sometimes applied in laboratory sample collection to obtain sufficient material quickly; however, this means that the morphology and composition of the samples may not reflect real-world emissions. Under extreme conditions, physical and chemical reactions may occur that would not normally take place, potentially leading to the formation of new compounds or altered particle properties. Therefore, it is important to distinguish whether BWPs were collected under simulated (including extreme) conditions or whether organisms were directly exposed to BWPs. In addition, a representative assessment of BWP toxicity requires careful consideration of whether the collection methodology and experimental conditions (realistic vs. extreme) adequately reflect the exposure scenarios encountered by organisms in urban or sparsely populated environments.

Taken together, these sampling techniques illustrate the methodological diversity in BWP research. Each method provides complementary information, but also introduces specific constraints related to representativeness, particle integrity, and relevance for human and ecological exposure scenarios. Careful selection and documentation of sampling techniques are therefore essential for reliable evaluation of BWP toxicity.

### *5.3. Current Methodological Limitations and Standardization Requirements*

Szabova et al. [63], highlighted several methodological challenges, including instrument calibration biases, particle losses in sampling lines, and inconsistent reporting of sampling parameters. Many instruments are calibrated using surrogate particles that do not reflect the morphology or composition of BWPs, leading to systematic uncertainties. The absence of harmonized protocols further complicates inter-study comparisons. Recent regulatory developments represent a critical step toward standardization by specifying controlled dynamometer testing, dilution systems, and isokinetic sampling requirements for BWP measurements.

The standardized methodology for measuring the number concentration of UFPs from the braking systems of light-duty vehicles, as defined by Global Technical Regulation No. 24 (GTR 24), is based on the Worldwide Harmonized Light Vehicles Test Procedure brake cycle. This test is conducted under laboratory conditions on a brake dynamometer and

simulates real-world driving. The cycle consists of multiple phases with varying braking intensity, speed, and thermal load. During testing, the brake system is enclosed within an airtight chamber that prevents the intrusion of untreated air and avoids mixing with the cooling air circulating around the brake assembly. The experimental setup includes continuous monitoring of both background particle concentration and brake temperature via thermocouples [93].

Figure 4 illustrates the arrangement comprising the minimum required subsystems for conducting brake emission tests with a brake dynamometer. The arrangement includes a climate control unit with a variable-flow fan (1), supplying the system with conditioned air. The conditioned air enters the brake enclosure (4), designed to accommodate the entire brake assembly under test. The brake dynamometer (6) enables and controls brake testing. The enclosure is directly connected to a sampling tunnel (7), at the end of which three (or four) sampling probes are mounted. These probes are used to collect aerosols from the tunnel for analysis by ultrafine and solid particle measurement devices. Downstream of the sampling plane, a flow measurement device (8) is positioned within the tunnel to monitor air velocity and flow rate [22,32].

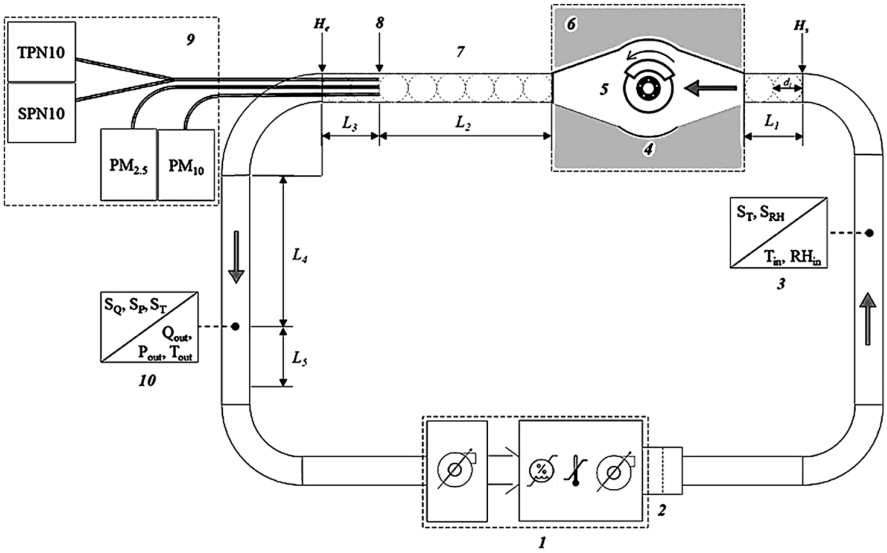


**Figure 4.** Schematic arrangement of the test stand: 1—climate control unit, 2—high-efficiency particulate air (HEPA) filter, 3—temperature and humidity sensors of the cooling air, 4—brake enclosure, 5—brake assembly connected to the brake dynamometer, 6—brake dynamometer (not illustrated, only indicated in grey), 7—sampling tunnel, 8—sampling plane with corresponding sampling probes, 9—measurement systems, 10—flow meter.

Figure 5 illustrates the schematic arrangement of sampling and measurement of UFPs. The test system must be capable of measuring the total particle number (TPN10) and the solid particle number (SPN10) of UFPs with a nominal electrical mobility diameter of approx. 10 nm and larger. The measurement system consists of the following components: sampling probe with an isokinetic sampling nozzle, transfer tube, cyclone, diluter, volatile particle remover, particle counter, and pump [22,32].

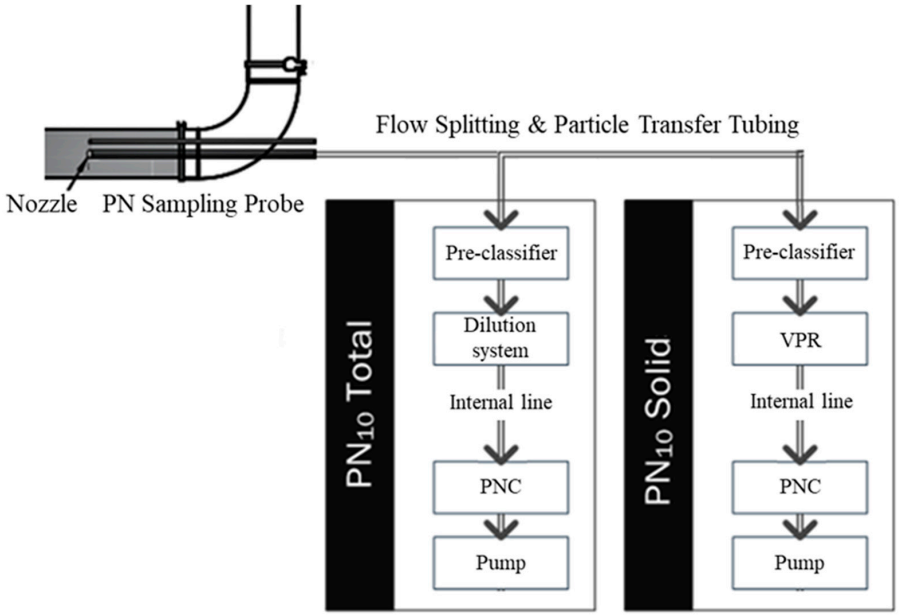


**Figure 5.** Schematic arrangement of the measurement system.

## 6. Ecotoxicological and Health Impacts of Brake Wear and Ultrafine Particles

Table 2 provides a comprehensive overview of current toxicological evidence related to BWPs and UFPs. Across a wide range of biological models, including human lung cell lines, lymphocytes, immune cells, aquatic organisms, plants, and mammals, the studies consistently highlight that BWPs can cause measurable biological effects. These effects vary significantly depending on particle size, chemical composition, brake pad formulation, and braking conditions during particle generation.

In certain cases [80,83,86,87,91], BWPs and UFPs were assessed only indirectly, while the primary focus was on emissions originating from general or traffic-related sources. Nevertheless, due to their relevance and informative value, these studies were included in the present analysis.

According to Puisney et al. [64], a proportionally higher number of toxicological studies have been conducted in vitro (outside a living organism, under artificial conditions) compared to in vivo studies (within a living organism, under natural biological conditions). This is because in vitro studies are technically simpler, less expensive to conduct, and impose fewer limitations regarding exposure routes and doses. This finding aligns with our own review of primary scientific articles.

**Table 2.** Summary of ecotoxicological and health impacts and experimental parameters of brake wear and ultrafine particles.

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Genotoxicity | Lymphocytes | Dynamometer | Brake pad model representing the average low-metallic composition of brakes available on the EU market. | Raman microspectroscopy, scanning electron microscopy (SEM) and TEM. | The genotoxic effect tested on human lymphocytes under various conditions showed an increased occurrence of micronuclei in binucleated cells exposed to BWPs in a culture medium, compared to lymphocytes exposed to the medium alone. | [27] |
| Cytotoxicity | Bronchial epithelial cells (Calu-3) | Dynamometer and mechanical collection | BWPs were emitted by the current European car fleet at a motor-vehicle testing center operating with light-duty vehicles. | Light scattering, X-ray fluorescence, thermogravimetric analysis, optical microscopy and SEM. | A short-term loss of cell viability associated with the formation of reactive oxygen species was confirmed, but with limited pro-inflammatory effects. | [64] |
| Cytotoxicity | Epithelial cells A549 | Dynamometer with an additional chamber simulating urban driving conditions | Commercial low-metallic and organic (asbestos-free) brake pads used in mid-range passenger cars. | The biological impact of BWP samples was studied during a 24 h exposure using a human 3D multi-cellular model consisting of human epithelial cells (A549) and human primary immune cells (macrophages and dendritic cells) mimicking the human epithelial tissue barrier. | Low-metallic BWPs did not induce any adverse biological effects in the in vitro lung cell model. However, BWPs from organic asbestos-free brake pads induced increased release of (pro)inflammatory mediators in the same in vitro system. | [65] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Cytotoxicity | Epithelial cells A549 | A model passenger vehicle (Renault Laguna 2.0) was placed on a lift in a garage and one front wheel was removed. An exposure chamber was then installed around the axle and brake system, with all sides carefully sealed with aluminum foil. | Unspecified | The lung cells cultured at the air-liquid interface were exposed to BWPs emitted during two typical braking modes ("full stop" and "normal deceleration"). | The particle number was significantly higher for "full stop 8×", i.e., $1130 \times 10^3$ #/cm$^3$ (SD $361 \times 10^3$) than for all the other braking behaviours. The concentrations of metals and carbon compounds in brake emissions increased with repeated braking and more intensive "full-stop" maneuvers. | [82] |
| Cytotoxicity | Lung adenocarcinoma cell line A549 | Dynamometer | Particles emitted during brake tests were generated by four different combinations of brake discs and pads with varying Cu content. | Lung cells were exposed to the BWPs ($PM_{2.5}$). The analyte content was measured in the friction brake materials using X-ray fluorescence (XRF) and in the $PM_{2.5}$ collected during braking tests using SEM and X-ray spectroscopy. | BWPs with the lowest Cu content caused no significant biological changes in lung cells compared to control samples, except for ROS generation and IL6 gene expression. In contrast, $PM_{2.5}$ with higher Cu content induced cytotoxicity proportional to the Cu concentration. | [66] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Cytotoxicity | Epithelial cells A549 | Dynamometer | Various types of brake pads including passenger cars and trucks | $PM_{10}$ extracts obtained from the brake samples were diluted to a final concentration of 150 ng/mL, except for samples derived from low-carbon and high-carbon brake pads during high-temperature braking cycles, where a maximum concentration of 400 ng/mL was achieved. | When comparing the results of the analysis of cells exposed to all PM extracts with the control group, no statistically significant difference was found. | [67] |
| Cytotoxicity | Alveolar adenocarcinoma cell line A549 | Mechanically collected from the drum brakes. | Drum brakes of seven passenger vehicles with an average age of 10 years and an average mileage of approx. 51,732 km. | The BWP samples were subjected to microscopic and elemental analysis. ROS formation was assessed using both acellular and cellular assays with the 2′,7′-dichlorodihydrofluorescein fluorescent probe and a cell-free fluorescence test. | ROS were detected using a cell-free fluorescent test, while the exposed cells were not significantly activated by the concentrations used. The fact that samples from older brakes have a lower oxidative stress potential than new brakes may be related to the highly oxidized or aged state of these materials, as well as their larger size and smaller reactive surface area. | [75] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Cytotoxicity | Lung adenocarcinoma cell line A549 | Pin-on-disc tribometer | 100 nm $Fe_2O_3$ UFPs (IOUFPs; Sigma-Aldrich, Taufkirchen, Germany); ceramic brake pads commonly applied in passenger vehicles in the Baltic region; ≥250 nm ceramic–ceramic wear particles produced by rubbing ceramic brake pad samples against each other; and ≤100 nm ceramic–steel wear particles generated by rubbing a ceramic brake pad sample against an S235JR steel disc. | Oxidative stress was evaluated by determining the levels of ROS, intracellular reduced glutathione, and malondialdehyde following exposure to the aforementioned particles at concentrations ranging from 10 to 80 µg/mL. Cells were exposed for durations of 4 and 24 h, respectively. | Both ceramic and ceramic-steel BWPs increased lipid peroxidation, but A549 cells showed lower sensitivity compared to Chinese hamster cells (*Cricetulus griseus*). | [94] |
| Cytotoxicity | Bronchial epithelial cell line BEAS-2B | Unspecified | Brake pads linings (type unknown) | Particle size was assessed using single particle extinction and scattering (SPES) after dispersing, sonicating, and briefly settling dust samples in water. Sizes were calculated from SPES-derived refractive index data. | All tested samples showed a genotoxic effect on BEAS-2B cells, even without inducing oxidative stress. BWPs showed to be the most cytotoxic compared to the other samples tested, which can be attributed to the high concentration of heavy metals, especially Fe and Cd. | [77] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Cytotoxicity | Bronchial epithelial cells (Calu-3) | Mechanically collected from the drum brakes—UFPs were monitored, as described and characterized in a previous study [68]. | Drum brakes of seven passenger vehicles with an average age of 10 years and an average mileage of approx. 51,732 km. | After sonication, UFPs were isolated by a combination of filtration and ultracentrifugation. This resulted in stable colloidal dispersions with a size peak in the 66 nm range. | Cu and Fe were identified as the predominant elements in the BWPs. The study showed that UFPs can cross the epithelial barrier in small quantities without compromising its integrity, indicating potential systemic effects due to their accumulation in secondary organs. | [68] |
| Cytotoxicity | Peripheral blood cells | Dynamometer | Various types of commercially available, low-metallic brake pads. | The collected particles were characterized by Raman microspectroscopy. | The BWPs studied were able to induce chromosomal damage under conditions simulating the natural environment of blood cells. | [69] |
| Cytotoxicity | Bronchial epithelial cells (Calu-3) | Mechanically collected | Collected from a bag filter in the ventilation duct of the testing plant in Sweden. The filter contained a year's accumulation of abrasion dust from drum brakes used in buses and trucks under conditions representative of urban driving and high-speed braking. | For BWP characterization X-ray fluorescence analysis, thermogravimetric analysis, SEM and optical microscopy were used. | The study showed that BWPs contained much higher levels of metals and metalloids than diesel exhaust particles (DEP). Despite this, both particle types caused similar effects in macrophages, including dose-dependent mitochondrial depolarisation ($p \leq 0.001$), increased secretion of IL-8, IL-10 and TNF-α ($p \leq 0.05$–$0.001$), and reduced phagocytic uptake of *S. aureus* ($p \leq 0.001$). | [74] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Respiratory tract | Lungs | - | Vehicles in a parking garage during the winter and summer campaign. | The samples were collected using time-integrated collection systems developed by the authors [79]. | Modeling of inhaled particle deposition revealed that 23.61% of UFPs are deposited in the pulmonary region of the lung, which corresponds to a dose of 10.67 µg in winter and 15.25 µg in summer for a normal 40 h work week. | [80] |
| Cytotoxicity | Alveolar epithelial cells type II (ATII) | BWPs collected in a laboratory setting using a high-volume cascade impactor onto foam filters. | Low-metallic; semi-metallic with limited copper; non-asbestos organic; and ceramic pads | BWPs in the 0.1–2.5 µm range was collected on polyurethane foam using a high-flow cascade impactor (0.9 $m^3$/min), extracted by sonication in methanol, and dried under nitrogen. | BWPs from organic and ceramic brake pads enriched with copper without asbestos showed the greatest oxidative stress, inflammation, and pseudohypoxic activation of hypoxia-inducible factor (a pathway associated with diseases related to air pollution, including cancer and pulmonary fibrosis), as well as disruption of metabolism and metal homeostasis compared to BWPs from low or semi-metallic brake pads and also PM from diesel exhaust. | [81] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Toxicity | Peripheral blood mononuclear cells | - | - | Twenty-nine healthy adults (20 men and 9 women) participated in a randomized two-factor crossover study with random exposure of participants to either unfiltered air containing particles in the UFP range or clean filtered air through a HEPA filter. | The study results suggest that UFPs, especially the 57 nm fraction, causes systemic oxidative stress with DNA damage without apparent compensatory activation of DNA repair mechanisms during 24 h exposure. | [85] |
| Carcinogenicity | $Sb_2S_3$ content in BWP samples | Dynamometer and mechanical collection | Dust from drum brakes came from 15 different trucks and three trailers and was taken with a clean plastic spoon from open brake drums in car repair shops during repair or replacement of brake pads. In five cases, the pads themselves were also changed and could be analyzed. | X-ray diffraction and selective solubility test | Selective solubility revealed that a significant amount of $Sb_2S_3$ oxidizes during the braking process, probably to form antimony oxide ($Sb_2O_3$), a suspected human carcinogen. According to this study, the use of antimony in friction brake materials should be considered a potential human carcinogenic risk and should be limited. | [76] |
| Carcinogenicity | $Sb_2S_3$ content in BWP samples | Mechanically collected | Used brake pads from various brands of passenger cars. Two samples of disc brake pads and three samples of brake pad wear debris accumulating on wheel rims were analyzed. | The composition of the BWPs was analyzed by combining SEM and energy dispersive spectroscopy (EDS) and X-ray absorption spectroscopy. | Brake pads contain extremely high levels of Sb, resulting from the use of $Sb_2S_3$. Assuming that all Sb emitted by cars comes from brakes, it can be assumed that the release occurs in the form of Sb(III) and Sb(V). | [78] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Toxicity | Cardiovascular system | - | - | Nine healthy, young non-smoking men working as members of a traffic highway patrol from 3 p.m. to midnight were monitored for four consecutive days during their service and the following night. | Fine particles from traffic can activate one or more signaling pathways that cause pro-inflammatory, prothrombotic, and hemolytic reactions in healthy young men. The long-term cardiovascular risk for police officers cannot be ruled out, especially considering the reported increase in myocardial infarction in professional drivers and the increase in mortality in people living near major road corridors. | [86] |
| Toxicity | Cardiovascular and respiratory system | Dust samples from five selected locations on the urban edge of a chosen agglomeration, which are heavily influenced by heavy and light trucks. | - | The device was designed by Amato et al. [83] and has been used in various studies, for example, by Alves et al. [84]. The collection methodology used was based on the procedure described by Amato et al. [83]. | The vehicle brake system was identified as a key urban source of thoracic particles and UFPs. | [83] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Toxicity | Cardiovascular and respiratory system | Contribution of UFP exposure from different types of transport (car, underground train, motorcycle and bus). | - | UFPs were measured with a P-Trak counter (TSI 8525). | The contribution of each transport mode to total daily integrated UFP exposure was 16.3–20.9% for car travel, 28.7% for motorcycle travel (which also includes exposure to BWPs, as indicated by the presence of Fe, Cu, and Ba), and 8.7% for subway travel. | [87] |
| Toxicity | Formation of hydroxyl radical (·OH) | Dynamometer | Non-asbestos organic, ceramic, and semi-metallic brake pads | BWPs were characterized using a combination of particle sizing instruments and cascade impactors. | ·OH as the most reactive form of ROS causing oxidative damage to lipids, proteins, and DNA. BWPs do not contain environmentally persistent free radicals, and all types of BWPs generate only hydroxyl radicals (·OH) in water. | [70] |
| Phytotoxicity | *Arabidopsis thaliana* | Dynamometer | The brake discs were not internally ventilated and had an outer diameter of 450 mm and a thickness of 356 mm and were manufactured out of grey cast iron (GJL-150). | To assess the physiological consequences of BWP exposure, transcriptomic data were obtained using RNA sequencing. | The UFP exposure triggered a wide range of gene expression changes, including disruptions in genes regulating iron and copper homeostasis. These findings indicate that metallic UFPs play a significant role in driving the physiological responses caused by BWPs. | [15] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Phytotoxicity | Cress seeds (*Lepidium sativum*) | Dynamometer | BWPs were generated from two types of brake pads, traditional phenol-based as a binder and new cement-based brake pads. | An acute phytotoxicity test was performed using cress seeds (*Lepidium sativum*) grown in artificial soil with increasing concentrations of BWPs from the two types of brake pads. | BWPs from simulated urban driving conditions affect the growth and damage of *Lepidium sativum* seedlings, especially in soil contaminated with high doses of these particles. The study results also suggest that particles from traditional phenol-based pads are more toxic than particles from innovative cement-based pads. | [71] |
| Aquatic toxicity | Green algae *Raphidocelis subcapitata* | Dynamometer | BWPs were generated from various types of commercially available, low-metallic brake pads. | Aquatic toxicity was determined using green algae (*Raphidocelis subcapitata*). Aquatic toxicity assessment was evaluated by using green algae *Raphidocelis subcapitata*. | The study results point to the potential for toxicity associated with airborne BWPs. The $EC_{50}$ value (50% inhibition of growth) was set at 85.46 g/L. According to the study authors, brake wear is probably much more toxic than ordinary road dust. | [69] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Zootoxicity | Bees—contamination of honey and pollen | The hive was surrounded by about 35 ha of uncultivated and cultivated lands of *Medicago sativa*, placed at approximately 150 m south of the main Italian highway, the Autostrada A1, and approximately 800 m away from the highway exit of the City. | Brake disc and brake pads. | A honey frame full of fresh honey was collected in August from the hive and the honey was prepared for a SEM/EDX analysis for the inorganic sediment contained in the sugar matrix, following a specifically modified protocol for melissopalynological analysis. | Pollen collected by foraging bees and the honey produced showed contamination with UFPs of iron oxides/hydroxides and baryte (a barium sulphate). Materials containing iron and baryte are common components of vehicle brake systems, and since another source of these substances was ruled out in the study location, the study authors assume that the contamination is primarily caused by vehicle brakes. | [91] |
| Zootoxicity | Springtails (*Collembola*) | The BWPs tested in this study were the same as those used in the study by Maiorana et al. [83], a full-scale dynamometer adapted to measure wear particle emission was used to generate and collect wear particles. | In this study, springtails (*Collembola*) were fed baker's yeast mixed with BWPs in two different concentrations, high and extremely low. | To test the toxicity of the BWPs, ten adults of *O. folsomi* per Petri dish per four replicates were used (four replicates per treatment and four replicates per control; 40 treatment and 40 control samples in total for each dose). | Chronic exposure to BWPs caused sublethal effects and histological changes even at low concentrations. SEM-EDX analysis detected chemical markers of BWPs in the gut and feces of the monitored springtail individuals (*Collembola*), while histological analysis found changes in the digestive and reproductive systems and the abdominal fat body at high concentrations. | [72] |

**Table 2.** *Cont.*

| Toxicological Domain | Endpoint | BWP Source or Generation Method | Brake Type or Composition | Analysis Method | Main Results/Conclusions | References |
|---|---|---|---|---|---|---|
| Zootoxicity | Female mice of the BALB/cOlaHsd strain | PM sampling was performed directly from the outlet of the sealed metal box. The normal supply of clean air to the sealed metal box was reduced to avoid particle losses during collection. | Four types of brake pads were used to collect $PM_{2.5}$ in an experimental setting under realistic simulated driving conditions at the Technical University of Ilmenau (Ilmenau, Germany). | The suspensions were then aerosolized for the purpose of inhalation exposure. Female mice were exposed by nose-only inhalation for 1.5, 3, or 6-h to a dose of up to 9 $mg/m^3$. | Exposure to BWPs from brakes induced inflammatory reactions in mice. However, a significant variability in the toxic effects of BWPs was found. | [79] |
| Zootoxicity | Male Wistar rats (Crl:Wi(Han)) | BWPs was generated from asbestos-containing drum brake shoes (1960s Chevrolet Impala, Bendix brand), which contained about 30% chrysotile asbestos but no amphibole asbestos. The confocal microscopy was performed by Rogers Imaging (Needham, MA, USA). | BWPs were generated from asbestos-containing drum brake shoes. | Groups of laboratory rats (Groups 1, 2, 3 and 4) were exposed for 6 h per day for 5 days to: Group 1 Filtered air (negative control Group); Group 2: Brake dust powder mixed with chrysotile 5R04; Group 3: Brake dust powder; Group 4: Crocidolite asbestos. | No notable pathological effects were detected at any time point in rats exposed to either BWPs alone or to the chrysotile/brake dust mixture. Inhalation of BWPs from chrysotile-containing brake drums does not cause lung pathology after short-term exposure. | [84] |
| Zootoxicity | Male Wistar rats | A semi-metallic brake lining was tested using a full-scale automotive brake AO4D dynamometer to study wear behavior, friction performance, and characteristics of released BWPs. | Collected wear debris, similar in composition to the friction layer, contained copper and iron oxides along with carbon-based materials. SEM and transmission electron microscopy were used to analyze morphology and size of BWPs. | Mutagenicity of brake wear extracts was tested using two bacterial assays (Ames test and SOS Chromotest) on chemical extracts prepared in dimethylsulfoxide. | Analysis showed that BWPs exhibited mutagenic potential in bacterial assays, suggesting the presence of indirect mutagens. Pulmonary toxicity tests in rats indicated an acute lung response to BWPs. The study concludes that further research is required to evaluate the long-term health risks of human exposure to BWPs. | [73] |

The main parameter used to assess BWP toxicity in current research is the formation of reactive oxygen species (ROS), which are commonly generated in response to environmental stressors, as concluded by Atonuje and Okpoghono [95]. Previous studies identified ROS generation in the presence of hydroxyl radicals (-OH) [70], Fe, Cu, and Cd [66,75,77]. According to Puisney-Dakhli et al. [68], the mucus layer covering bronchial epithelial cells plays a protective role by enabling the mucociliary self-clearing mechanism via ciliated cells. Its production must be tightly regulated to avoid excessive mucus formation, which could impair mucociliary activity and increase susceptibility to infections. Other studies assessed BWP toxicity through additional biological endpoints, including reduced phagocytic activity of monocytes [69], pro-inflammatory responses [48,64,65,74,79,81,82,84], and pro-thrombotic effects [86].

In addition to direct experimental measurements, modeling approaches were also employed for evaluation [45,49,80].

It must be acknowledged that not all identified ecotoxicity studies were directly linked to brake systems. Available epidemiological and experimental evidence points to the significant role of air pollution, particularly $PM_{2.5}$ and UFPs, in the development of a wide range of cardiovascular and neurodegenerative diseases. A cohort study of an extensive elderly population in Taiwan showed that long-term exposure to $PM_{2.5}$ increases the risk of Alzheimer's disease [88].

Cardiovascular effects have been confirmed in several observational studies. Data from Augsburg showed that short-term increases in particles within the 10–100 nm range were associated with a higher incidence of myocardial infarction, with risk increasing proportionally to particle surface concentration and decreasing particle diameter [89]. Similarly, in Helsinki, associations were observed between concentrations of $PM_{2.5}$, UFPs, and carbon monoxide with fatal strokes, but only during the warm season, likely reflecting seasonal variations in aerosol composition or exposure intensity [90].

Experimental studies provide mechanistic support for these epidemiological associations. In Los Angeles, UFPs were shown to induce oxidative stress through upregulation of heme oxygenase-1 and depletion of intracellular glutathione. These effects were directly linked to the high organic carbon and polycyclic aromatic hydrocarbon content of the particles, and UFPs also caused mitochondrial damage. Similar findings were confirmed by ROS formation assays using dithiothreitol, demonstrating a strong correlation between the chemical composition of UFPs and their oxidative potential [96].

In some in vivo studies, behavioral alterations in test organisms were also observed, such as changes in anxiety-related behavior in mouse offspring. These findings are valuable because they indicate potential impacts on the nervous system, motor function, and overall organismal health. However, behavioral outcomes alone do not provide a comprehensive toxicological profile and must be interpreted alongside biochemical and physiological endpoints, as behavior can be influenced by stress or trauma during substance administration. Minimizing such confounding factors requires appropriate habituation procedures, a stable environmental setting, and careful handling of test organisms [92].

Controlled clinical exposure studies further confirm the adverse effects of UFPs in susceptible human populations. In individuals with metabolic syndrome [44] and in patients with type 2 diabetes [51], inhalation of concentrated UFPs was associated with cardiovascular disturbances, including impaired heart rate variability. Similarly, short-term exposure of asthmatic subjects to freshly generated carbon particles [50] did not induce acute inflammatory exacerbations; however, subgroup analyses suggested potential long-term implications for disease progression. Children represent one of the most vulnerable groups, and a study conducted in an urban environment demonstrated that UFP exposure

increased the risk of asthma exacerbations, with the strongest effects observed in children regularly taking preventive medication [45].

Evidence related to microvascular impairment is also notable. Personal exposure monitoring showed that UFPs from ambient sources are associated with microvascular dysfunction, likely mediated by systemic inflammation [48]. Vascular disturbances linked to UFPs were similarly detected through measurements obtained at an urban monitoring station [46].

Model-based approaches provide further insight. A chemical transport model applied in California identified specific PM components associated with ischemic heart disease mortality, reinforcing the notion that particle composition, rather than mass concentration alone, plays a decisive role in toxicity [49].

Vanbrabant et al. [92] also demonstrated reproductive and developmental toxicity. In an experimental mouse study, postnatal exposure to UFPs resulted in behavioral alterations indicative of neurotoxic effects, suggesting that the consequences of UFP exposure may arise early in development and persist throughout later life.

Elemental analyses using ICP-MS and ICP-OES have consistently demonstrated a pronounced enrichment of several heavy metals—particularly Cd, Cr, Zn, and V—in UFPs compared to bulk dust collector material, with reported nano-to-bulk concentration ratios $\geq 2$ [97]. This enrichment indicates preferential partitioning of toxic metals into the UFP size fraction, which is of particular concern due to the enhanced mobility, bioavailability, and respiratory deposition efficiency of these particles. According to the same study, road-derived UFPs contained approximately twofold higher concentrations of Cd, Zn, and V, and up to ninefold higher concentrations of Cr, compared with UFPs sampled from local streets [97]. The highest median Zn concentrations were observed on urban expressways, likely reflecting increased traffic density, the presence of light- and heavy-duty vehicles, and higher driving speeds, all of which intensify brake wear processes.

Several toxic and neurotoxic metals, including Mn, Zn, Pb, Cu, Ni, and Ba, are known to adsorb onto particulate matter surfaces and may substantially contribute to adverse neurological outcomes associated with PM exposure. Epidemiological evidence supports a negative association between Mn bound to PM and neurodevelopmental outcomes in children, as well as impaired motor function in elderly populations [98]. These findings highlight the relevance of metal-rich UFPs from non-exhaust sources, such as brake wear, as an underappreciated contributor to traffic-related health risks.

Advanced analytical approaches, including ICP-MS/MS combined with SEM/EDX, have been employed to characterize the metal content of ultrafine fractions, revealing the presence of Cu, Cr, Mn, and Zn in addition to Fe [99]. The elemental fingerprints observed in these studies strongly suggest a substantial contribution from brake disc wear, underscoring the importance of braking system components as dominant sources of metallic UFPs. Similarly, SEM–EDAX analyses conducted on brake pad materials reported no detectable levels of certain carcinogenic elements, such as Cd, Pb, and Sb, in the investigated samples [100]. While this finding may indicate improvements in brake pad formulations, it does not preclude the enrichment of other potentially harmful metals in emitted UFPs during braking.

Further insight into brake pad composition has been provided by FESEM coupled with EDX mapping, which revealed that commercial brake pads are primarily composed of Ti, C, and Cu, with secondary constituents including Si, K, Fe, and Mg [101]. These compositional characteristics are consistent with observed UFP emissions enriched in metallic elements, particularly Cu and Fe, which are commonly used as friction modifiers and structural components.

Real-world exposure studies have also confirmed the presence of brake-related metals in ambient UFPs. Mobile monitoring campaigns conducted near Antwerp, Belgium, reported commuter exposure to UFP-associated metals such as Cu (26–46 ng/m$^3$), Cr (47–63 ng/m$^3$), Mo (3.83 ng/m$^3$), and Ni (4–21 ng/m$^3$), as determined by HR-ICP-MS [102]. These concentrations underscore the relevance of non-exhaust emissions as a continuous source of metal-bearing UFPs in urban environments.

Experimental investigations using sintered brake pads further demonstrated that Fe and Cu dominate the elemental composition of particles collected on filters, accounting on average for approximately 25% and 22% of the particle mass, respectively [103]. Notably, the mass fraction of Fe decreased with increasing braking energy, whereas Cu exhibited the opposite trend, suggesting different wear and emission mechanisms for individual metals. Chromium was also detected as a minor but non-negligible constituent, with an average contribution of 2.9% [103].

A comprehensive methodological study combining spectroscopic techniques, thermogravimetric analysis (TGA), chromatography, and morphological and elemental analyses confirmed that BWPs are predominantly metallic in nature [104]. TGA results indicated that more than 90% of the sample mass consisted of metals, with Fe representing the largest fraction (50–80%), primarily in oxidized forms. Other metals, including Zn, Al, Mg, Si, S, Sn, and Mn, were present at lower concentrations (approximately 1–2% each), findings that were corroborated by X-ray diffraction (XRD) analysis [8]. Collectively, these studies demonstrate that brake-generated UFPs are highly enriched in heavy metals, reinforcing their potential significance for environmental exposure and human health risk assessment.

Collectively, these findings indicate that UFPs represent a significant health hazard. The convergence of epidemiological observations and mechanistic evidence supports the conclusion that UFPs exert their effects primarily through oxidative stress, inflammation, and mitochondrial dysfunction. Vulnerable populations, including children, older adults, and individuals with pre-existing metabolic or respiratory conditions, are particularly at risk.

## 7. Concentration Metrics and Size Distributions of Brake Wear and Ultrafine Particles

Only a subset of the above mentioned studies in Table 2 quantified mass or particles number concentration directly, many toxicological investigations instead expressed dose as surface mass concentration (μg/cm$^2$) or suspension concentration (μg/mL). Therefore, the mass and particles number concentration of BWPs were not discussed in detail in the previous text.

It is important to note that the EURO 7 emission standard represents the most comprehensive regulatory framework to date aimed at reducing pollutant emissions from road transport; however, it remains limited in its treatment of non-exhaust particle emissions. While EURO 7 introduces, for the first time, a requirement to measure and control brake abrasion mass concentration of $PM_{10}$ from light-duty vehicles, it does not include limits or measurement protocols for brake-derived particle number concentrations, particularly in the ultrafine range. This omission is notable given that toxicological studies indicate that the ultrafine fraction of BWPs, best represented by particle number–based metrics, has the potential to cause adverse health effects. Therefore, concentration metrics are reported separately, which is essential for scientific clarity, methodological accuracy, and meaningful interpretation of toxicological outcomes.

Across the analyzed studies, BWPs were characterized under a variety of experimental conditions. The substantial variability in exposure metrics limits direct quantitative comparison; however, collectively, the available studies encompass a wide range of traffic-

relevant BWP concentrations, spanning approx. $10^3$–$10^5$ #/cm$^3$ and from a few μg/m$^3$ to several mg/m$^3$ in mass. The following text presents examples from available studies that evaluated particle number and mass concentrations, as well as particle size distributions.

Gasser et al. [82], provided one of the most detailed characterizations of BWPs in terms of mass concentration, particle numbers, and size distribution, offering important context for interpreting toxicological outcomes. During dynamometer experiments, the authors observed short-term airborne mass concentrations ranging from approximately 2.0 to 4.5 mg/m$^3$, depending on braking intensity, with the highest values occurring during repeated full-stop events. Elemental analysis confirmed the dominance of transition metals, particularly Fe, which alone contributed between 1.0 and 3.8 mg/m$^3$ to the total particulate mass, while Cu (33–163 μg/m$^3$), Mn (9–28 μg/m$^3$), organic carbon (120–258 μg/m$^3$) and elemental carbon (12–35 μg/m$^3$) represented smaller but significant fractions. Parallel number-based measurements demonstrated that the aerosol was strongly enriched in ultrafine fraction: mean number concentrations increased from approx. $6.5 \times 10^5$ #/cm$^3$ during constant-speed operation to approx. $1.1 \times 10^6$ #/cm$^3$ during intensive braking. Particle number size distributions exhibited a dominant mode at approx. 100 nm and a secondary accumulation-mode peak near 300–400 nm, indicating that freshly generated BWPs are predominantly ultrafine. Deposition analysis further supported this conclusion, with $9.3 \times 10^6$ to $3.3 \times 10^7$ #/cm$^3$ depositing onto TEM grids at the cell exposure location. Collectively, these data confirm that real-time brake wear generation leads to transient but extremely high UFP-rich aerosols in the immediate microenvironment. Such conditions provide essential mechanistic insight into why BWPs, and especially their ultrafine fractions, exert strong oxidative and proinflammatory effects in vitro, as the number-based and surface-related particle metrics far exceed what is reflected by mass concentration alone.

Richtmann et al. [15] measured brake-related UFP number concentrations of approx. from $6.5 \times 10^3$ to $2.7 \times 10^4$ #/cm$^3$ at brake test bench. Particles were collected using an impactor (ELPI+, DEKATI) and the ultrafine fractions were analyzed with spICP-MS. The particle diameter range assessed was 6–100 nm, with median diameters of 10–105 nm depending on braking severity in city, cross-country and highway. Consistent with the findings of Gasser et al. [82], this study also demonstrated that the measured BWPs were dominated by UFPs.

In contrast, Selley et al. [74], used SEM to characterize BWPs and reported particle diameters ranging from 0.2 to 22.9 μm, with a median diameter of 2.4 μm. Moreover, dynamic light scattering measurements indicated a mean BWP diameter of approx. 0.6 μm. These values lie slightly above the size range typically associated with the ultrafine fraction.

Kazimirova et al. [25], directly observed UFP presence in airborne BWPs, with particle diameters as small as 20 nm, confirmed by TEM. However, no numerical size distributions or modal diameters were provided beyond this. In addition, study does not contain any data on mass or number concentrations of the samples.

The study by Zhao et al. [75], provided detailed structural characterization of aged BWPs, but likewise did not report airborne mass or particles number concentration. Nevertheless, TEM analysis revealed that the material contained abundant nanoscale primary particles as small as 10–15 nm, aggregated into larger clusters. These nanosized features were consistent with the authors' earlier observations during real-time braking, in which 74% of emitted particles were nanoparticles and number-size distributions exhibited dominant modes at 60–100 nm, with an additional < 30 nm nuclei-mode appearing only under high thermal load. The presence of similarly sized nanoscale structures in the aged debris indicates that ultrafine constituents persist after mechanical abrasion and environmental transformation. Although no aerosol concentration metrics were measured directly, the consistent detection of 10–100 nm particles highlights the intrinsic UFP richness of brake-wear

material and reinforces that nanoscale components remain a relevant exposure fraction even after aggregation and aging.

Maiorana et al. [71], investigated the influence of BWPs from traditional phenolicresin-based pads and innovative cement-based pads on cress seed germination, root growth and damage. Information on BWP size was obtained using SPES. The BWPs generated from both pads exhibited similar dimensions and size distributions, and the particle number mean diameter (0.56 and 0.57 μm).

Gerlofs-Nijland et al. [79], focused on controlled inhalation of BWPs collected from four different types of brake pads in experimental setting under realistic simulated driving conditions inside a large air-conditioned test-cabin. They used mass concentrations of approx. 3.0, 8.0 and 9.0 mg/m$^3$ for low, medium and high exposures, during repeated 6 h inhalation sessions. Furthermore, the experiment included measurements of BWPs number concentrations, which ranged from about $1.0 \times 10^5$ to $1.5 \times 10^5$ #/cm$^3$, with count median diameters between 58 and 73 nm for the ultrafine fraction.

The comparative study by Bernstein et al. [84], evaluated brake pads manufactured with chrysotile and crocidolite asbestos. BWPs were generated by grinding the brake shoes using a commercial grinder equipped with a dust-collection system. For the toxicological rat exposures, mean particle number concentrations ranged from approx. 600 to 3000 #/cm$^3$, with geometric mean diameters between 0.17 and 0.34 μm. The corresponding gravimetric concentrations ranged from approx. 1.5 to 6.3 mg/m$^3$.

Kukutschová et al. [73], demonstrated that particle size measurements are strongly influenced by aggregation and sample handling. SEM imaging revealed that BWPs consisted of a wide range of morphologies, including large steel chips released at elevated temperatures due to matrix degradation, and granular particulate matter prone to forming aggregates. SEM-derived particle sizes averaged approx. 200 μm. They concluded that many large particles were in fact agglomerates of smaller fragments, while the smallest particles were likely underrepresented because they adhere to one another during suspension preparation. The results emphasize that BWPs readily coagulate or fragment depending on suspension conditions, underscoring the need for careful interpretation of size-distribution data when evaluating wear-particle characteristics.

Mass concentrations reported by Vaze et al. [80], indicate that the analyzed PM consisted predominantly of fine and ultrafine fractions, with measured levels in the range of tens of μg/m$^3$. By contrast, the number concentrations reached values of $10^4$–$10^5$ #/cm$^3$, showing that the particle count is highly sensitive to the nanosized fraction. The study demonstrated that although coarse particles contributed appreciably to the total mass, they accounted for only a negligible fraction of the particle number concentration. However, this study examined a broader aspect of vehicular contributions to particulate emissions, extending beyond BWPs alone.

## 8. Study Limitations and Future Perspectives

Despite substantial progress in understanding the physicochemical characteristics and toxicological relevance of BWPs, significant unresolved issues persist with respect to the UFPs. One of the principal knowledge gaps concerns the incomplete mechanistic understanding of UFP formation and behavior during braking under real-world conditions. Although both mechanical abrasion and thermally induced degradation of brake materials are recognized as dominant generation pathways, their relative contributions remain insufficiently quantified across different braking regimes, vehicle classes, and brake system designs. In particular, the conditions governing the transition from mechanically generated nanoscale fragments to thermally driven nucleation of secondary UFPs through vaporization–condensation processes are poorly constrained.

Another unresolved issue is the lack of harmonized and standardized metrics for characterizing brake-derived UFPs. Most existing regulatory frameworks and experimental datasets rely predominantly on mass-based indicators, which are inherently unsuitable for characterizing UFPs, as these particles dominate number concentration, surface area, and chemical reactivity while contributing negligibly to total particulate mass. Particle number concentration, surface area, oxidative potential, and metal bioavailability are inconsistently reported across analyzed studies, while methodological differences in sampling and measurement techniques further complicate cross-study comparability. Sampling losses, particle aggregation, and instrument detection limits introduce additional uncertainty and may result in systematic underestimation of BWP and UFP effects. Moreover, epidemiological evidence linking health effects specifically to brake-derived emissions remains limited, largely because brake wear emissions are commonly grouped with other traffic-related ultrafine particle sources.

The absence of long-term cohort studies with robust source apportionment and the lack of specific chemical or morphological tracers for brake-derived UFPs hinder accurate risk attribution. This limitation is compounded by the scarcity of monitoring of UFPs in real-world urban conditions, which restricts the availability of exposure data necessary for large-scale epidemiological assessments.

Further uncertainty arises from the limited understanding of chronic and low-dose exposure effects. Most toxicological studies employ short-term or acute exposure scenarios, frequently at elevated concentrations, to elicit measurable biological responses. However, real-world exposure to UFPs is typically chronic, intermittent, and spatially heterogeneous. The long-term consequences of repeated low-level exposure, cumulative particle retention in lung tissue, potential systemic translocation, and delayed effects on cardiovascular, immune, and nervous systems remain insufficiently characterized. These uncertainties are particularly relevant for vulnerable populations, including children, the elderly, individuals with pre-existing cardiopulmonary conditions, and occupationally exposed groups such as professional drivers.

The environmental fate and transformation of UFPs also remain incompletely understood. After emission, UFPs undergo aggregation, deposition, resuspension, and chemical aging processes that may substantially alter their bioavailability and toxicity. Oxidative transformation, interaction with secondary organic aerosols, and changes in metal speciation during atmospheric aging can modify particle reactivity and biological effects. Similarly, pathways governing the transfer of UFPs into soils, surface waters, vegetation, and food chains, as well as their persistence and bioaccumulation potential, are only partially elucidated.

From a regulatory perspective, the absence of particle-number-based limits for brake wear emissions represents a critical gap. Without regulatory requirements for UFP monitoring and control, mitigation strategies remain largely voluntary and inconsistent across manufacturers and jurisdictions.

Future research should therefore prioritize the development of standardized methodologies for the measurement and reporting of UFPs, including harmonized particle number–based metrics and complementary indicators such as surface area and oxidative potential.

Long-term epidemiological studies with improved source apportionment are essential to clarify the specific health impacts of brake-derived UFPs. Mechanistic toxicological studies focusing on chronic, low-dose exposure scenarios are needed to better reflect real-world conditions and to identify cumulative and systemic effects.

Risk mitigation strategies should adopt a multipronged approach. At the technological level, optimization of brake pad and disc formulations offers substantial potential for reducing UFPs, particularly through the reduction or substitution of highly reactive metals

such as Cu, antimony, and cadmium, and through the use of alternative binders with lower thermal degradation potential. Advances in brake system design, including improved heat dissipation, surface coatings, and regenerative braking technologies, especially in electric and hybrid vehicles, can further reduce frictional load and particle generation. In addition, enclosing the disc brake within a shroud and integrating particle capture can substantially reduce BWP emission by limiting direct dispersion and directing particle-laden airflow toward a defined filtration or extraction point. Current industrial development focuses on passive filters mounted near the caliper and active suction and filtration concepts triggered during braking, indicating that source-adjacent capture is technically feasible and moving toward practical implementation.

At the policy level, incorporation of particle-number-based limits for UFPs into future regulatory frameworks would represent a critical step toward effective control of non-exhaust vehicle emissions. Finally, urban planning and traffic management strategies, including coordinated signal control (green wave) and speed regulation, may reduce braking frequency and intensity and thereby indirectly lower UFP exposure in densely populated areas.

Addressing these unresolved issues through coordinated scientific, technological, and regulatory efforts is essential to improve exposure assessment, reduce health and environmental risks, and ensure that future mitigation strategies adequately reflect the role of UFPs in vehicle emissions.

## 9. Conclusions

The available evidence demonstrates that BWPs represent a substantial source of toxicologically relevant non-exhaust emissions. Across a broad range of in vitro, in vivo, and ecotoxicological models, BWPs consistently induce adverse biological responses, the magnitude and nature of which are strongly governed by particle size, chemical composition, and braking conditions.

UFPs and metal-enriched fractions, particularly those containing Cu and Fe, exhibit elevated oxidative and inflammatory potential. These particles induce oxidative stress, mitochondrial dysfunction, cytotoxicity, and genotoxic effects in lung epithelial and immune cells, while respiratory deposition modelling confirms efficient penetration of UFPs into the alveolar region. Evidence of cardiovascular alterations and oxidative DNA damage further indicates that brake-derived UFPs may contribute to systemic health effects beyond the respiratory tract.

Environmental studies corroborate these findings, showing adverse impacts of BWPs on aquatic organisms, plants, soil invertebrates and higher trophic levels, with indications of bioaccumulation and transfer of BWPs within terrestrial and aquatic ecosystems. Variability in toxicological outcomes is consistently linked to brake pad formulation and braking severity, with metal-rich materials and high-intensity braking producing particles of greater reactivity and biological potency.

Although BWPs encompass a wide particle size spectrum, emissions are dominated by UFPs in terms of particle number, whereas mass concentrations are governed by larger fine and coarse particles. This size-dependent discrepancy underscores a fundamental limitation of mass-based emission metrics in capturing the most toxicologically relevant fraction of brake emissions. Consequently, the findings highlight the need for standardized particle number-based characterization and regulatory control of BWPs, which remains insufficiently addressed in current legislative frameworks.

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